LLMs in Bioinformatics

Student pre-work

November 14-15, 2026

- Check that you can open Colab and run one Python cell.
- Local Python / VS Code setup is optional.
- Bring your laptop, charger, and questions.

Before the workshop



- Bring a laptop and charger; check your browser and network.
- Sign in to Google Colab with a Google account.
- Run the small test cell shown on the next slide.
- No paid plan, GPU, GitHub account, or API key is required for this pre-work.
- If setup fails, bring the error with private information removed.

Open Colab and test one cell



- Open colab.research.google.com and sign in.
- Create a new notebook and run the code below.
- Check the output beneath the code cell.

```
print("Hello, bioinformatics!")
```

Save your notebook. Runtime files and installed packages are temporary.

Use the slides and notebook together



- Read the slides to frame the biological question and method.
- Open the instructor-provided notebook for that activity.
- Save a copy in Drive before editing a shared workshop notebook.
- Run cells in order and inspect their outputs.
- Return to the slides to interpret the results.

The organizers will provide the final workshop notebook links.

Optional local Python setup



- Install Python 3 and VS Code.
- Install Microsoft's Python extension.
- For local notebooks, add the Jupyter extension.
- Create a .venv and select its interpreter / kernel.

Official VS Code tutorial, captured November 9, 2026.

Follow the student guide. Install lab packages only when final requirements are confirmed.

Keep research data safe



- Use only public or synthetic workshop data.
- Do not upload identifiable, clinical, confidential, unpublished, or restricted data to personal cloud notebooks or AI tools.
- Never put passwords, API keys, or access tokens into notebooks or prompts.
- Share setup errors with private details removed; a teaching assistant can help.

Optional AI access programs

GitHub Copilot Student

- Eligible enrolled students can apply through GitHub Education.
- Provide current enrollment proof, then activate Copilot after approval.
- See the guide for official eligibility and application links.

Claude for research labs

- Anthropic's scientist access program is PI-led.
- An eligible academic / nonprofit PI verifies first and may add lab members.
- Ask your PI and check current terms using the official links in the guide.

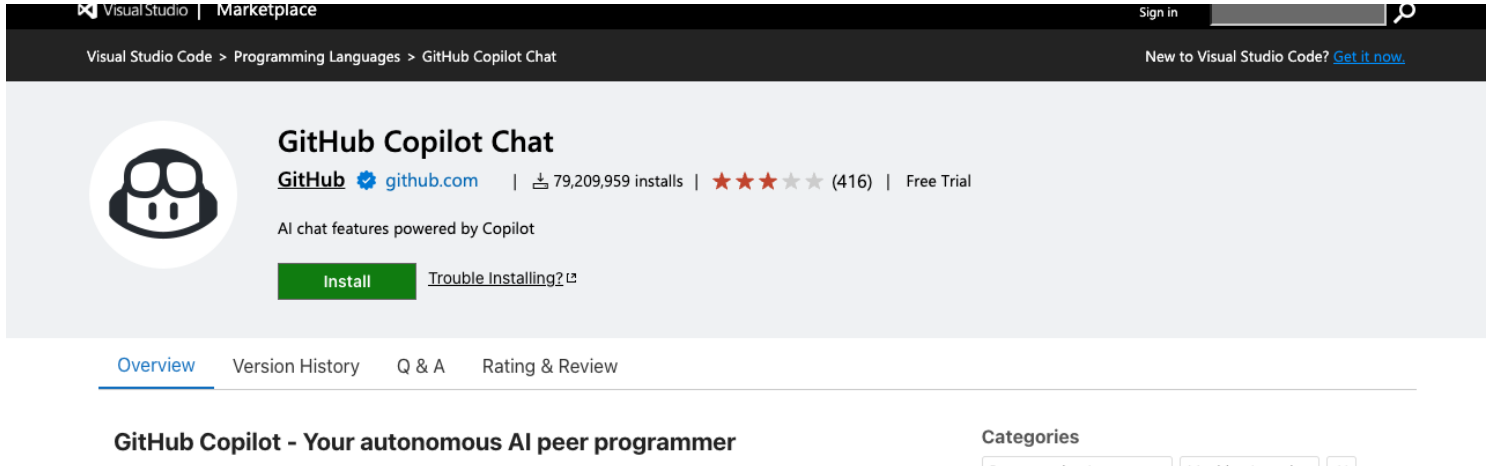
Neither service is required for this pre-work. Do not buy a plan for it.

Create a GitHub account (optional)

- Reuse your existing personal account, or open github.com/signup.
- Follow the signup prompts; choose a username and verify your email.
- Use a free account; no paid plan is needed for this optional setup.
- Enable two-factor authentication and keep recovery codes private.
- Check you can sign in. Education verification is a separate step.

GitHub: creating a personal account (checked November 9, 2026)

GitHub Copilot in VS Code (optional)



Official Marketplace page, captured November 9, 2026. Browser view, not an installed-extension screen.

Publisher: GitHub

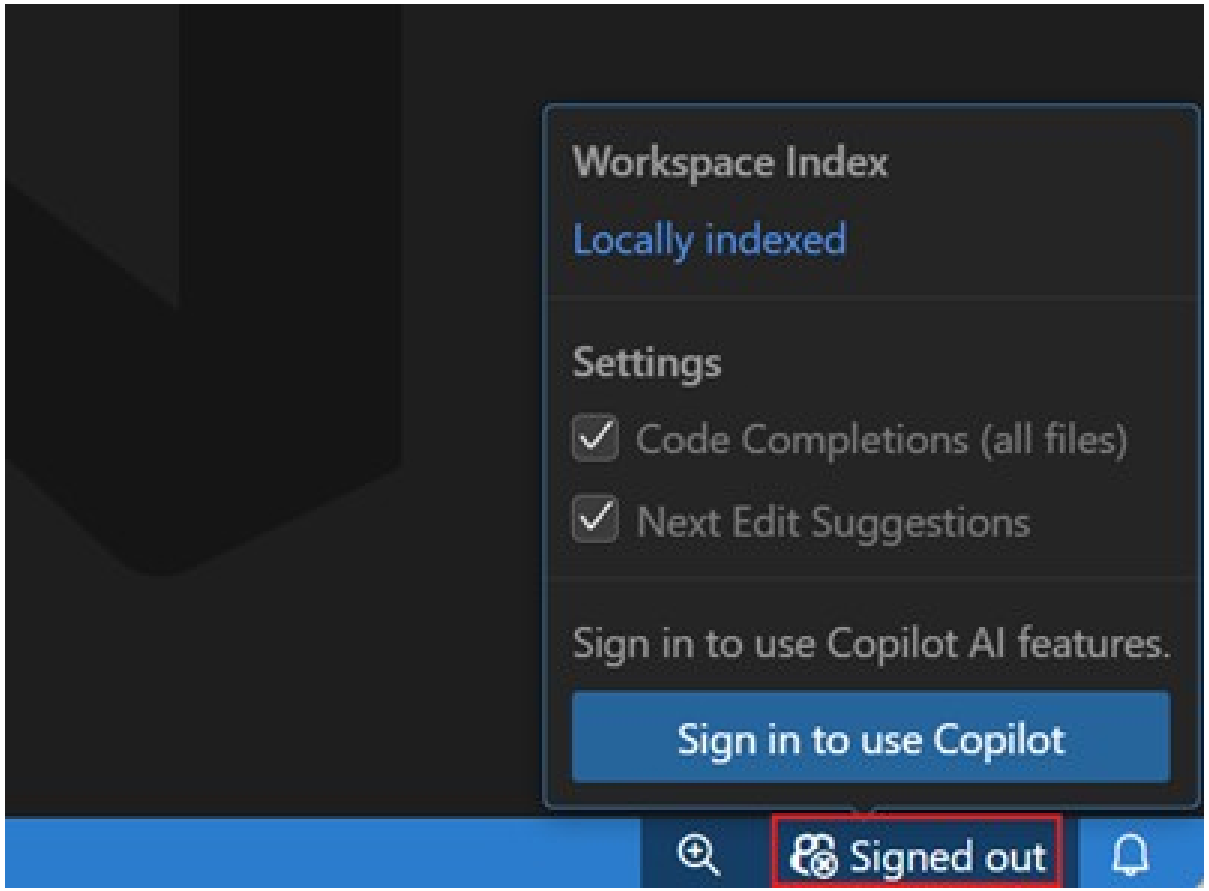
Extension ID: GitHub.copilot-chat

Installation and Copilot plan activation are separate.

- 1. Update VS Code; open View > Extensions.
- 2. Search GitHub Copilot and verify the publisher / ID.
- 3. Choose Install. From the web page, allow it to open VS Code, then confirm there.
- 4. Sign in as shown next. Existing installs may show Manage instead.

Official extension listing; setup guide is linked in the student handout (checked November 9, 2026)

Copilot: sign in after installation

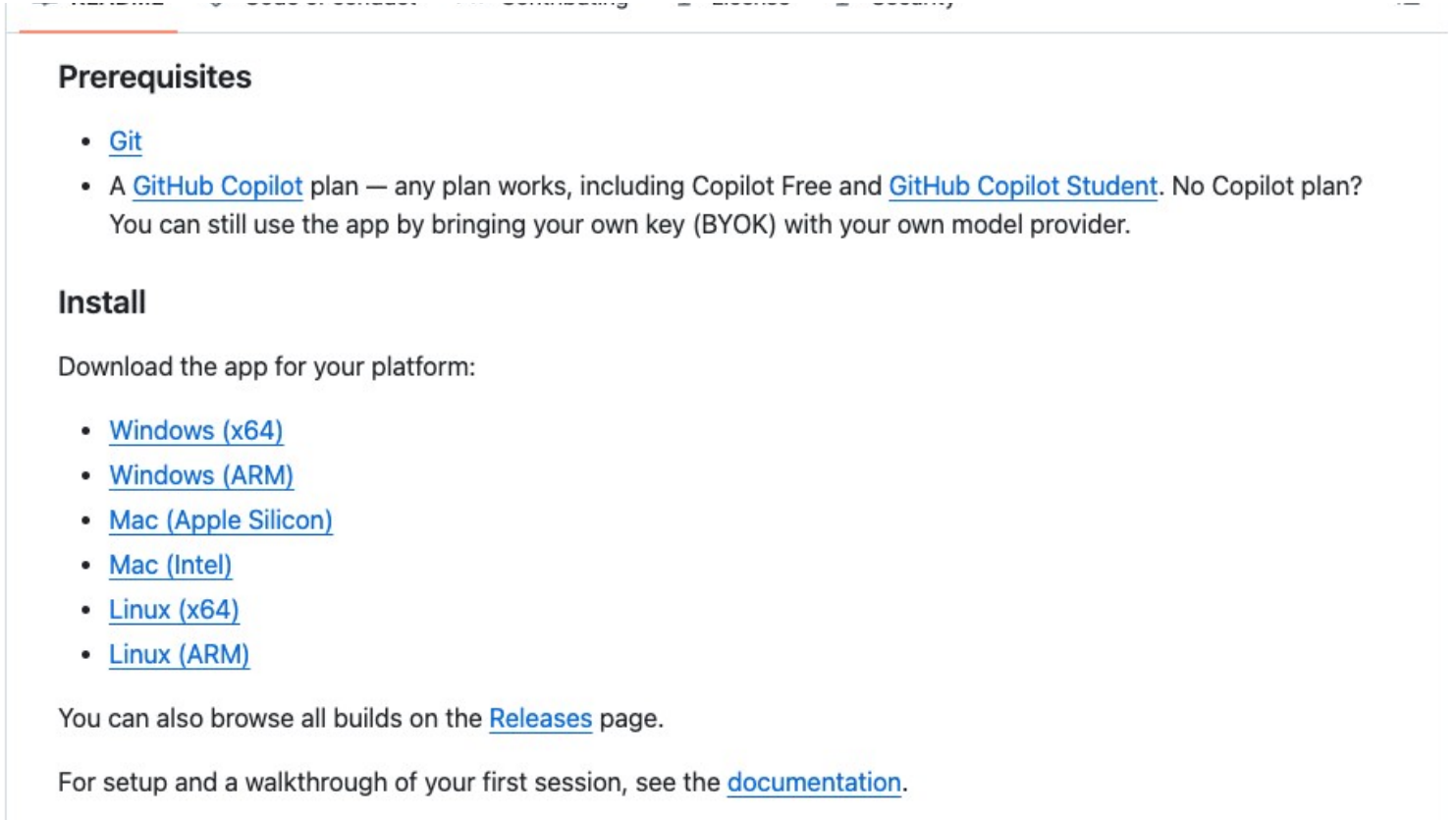


Official VS Code documentation UI example, captured November 9, 2026. Your menu may differ.

- 1. In VS Code, open the Copilot status/menu.
- 2. Choose Sign in to use Copilot; finish GitHub authorization in the browser.
- 3. Use your Free, Student, or existing institutional account; verify the plan.
- 4. Test in a scratch folder: Explain hello.py. Review the answer.

Official VS Code Copilot setup and sign-in (checked November 9, 2026)

GitHub Copilot desktop app (optional)



Prerequisites

- Git
- A GitHub Copilot plan — any plan works, including Copilot Free and GitHub Copilot Student. No Copilot plan? You can still use the app by bringing your own key (BYOK) with your own model provider.

Install

Download the app for your platform:

- Windows (x64)
- Windows (ARM)
- Mac (Apple Silicon)
- Mac (Intel)
- Linux (x64)
- Linux (ARM)

You can also browse all builds on the Releases page.

For setup and a walkthrough of your first session, see the documentation.

Official github/app prerequisites and download links. Browser capture: November 9, 2026; no installer was run.

- 1. Install Git. Visit github.com/github/app.
- 2. Select your OS and processor (Apple Silicon / Intel or x64 / ARM).
- 3. Run the downloaded platform installer, open Copilot, and sign in to GitHub.
- 4. Use existing/Free/Student access; open a practice folder and start Interactive mode.

This is the GitHub Copilot app, not GitHub Desktop or Microsoft Copilot.

GitHub Copilot app: installation and first session (checked November 9, 2026)

Freebuff: privacy warning

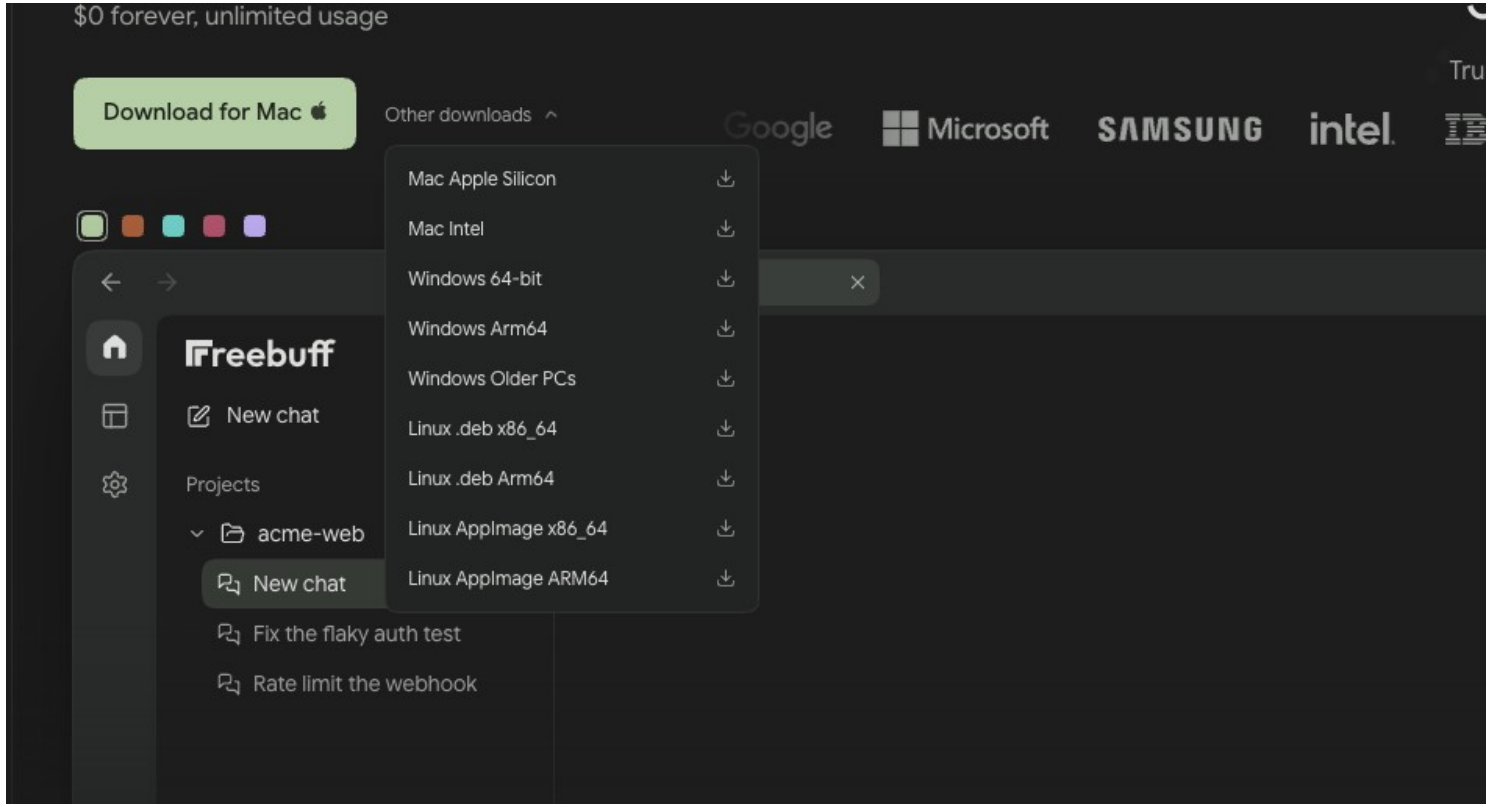
Do not use sensitive or unpublished research data.

- Prompts and project content may be processed by Freebuff and model providers.
- Project/prompt data may be used to personalize advertising.
- AI-training-labeled models/features may allow data licensing or sale; this is not a blanket rule for every mode.
- Desktop/CLI agents can read beyond the project; a local app is not proof of privacy.

Workshop practice: public/synthetic data only. Skip Freebuff if unsure.

Freebuff policy: updated November 8; effective November 12, 2026 (checked November 9, 2026)

Install Freebuff (optional)



Freebuff website: Other downloads expanded on macOS. Captured November 9, 2026. Background app preview is a vendor demo.

- 1. Arrange an isolated OS account/environment first; otherwise skip.
- 2. Visit freebuff.com. Open Other downloads if the suggested OS/CPU is wrong.
- 3. Choose the matching desktop installer. Read live terms/privacy before installing.
- 4. Install and sign in only in the isolated environment; do not connect research data.

[Freebuff official desktop downloads (checked November 9, 2026)]

Use Freebuff safely

- Create hello.py with print("Hello, bioinformatics!") in a disposable practice folder.
- Use only public/synthetic files and keep an unchanged backup.
- Agents may access files beyond a project folder; it is not a sandbox.
- Try: Explain hello.py. Do not edit files or run commands.
- Review any proposed edits and commands; check the outputs yourself.

Read data handling and advertising terms. A local app is not necessarily local-only.

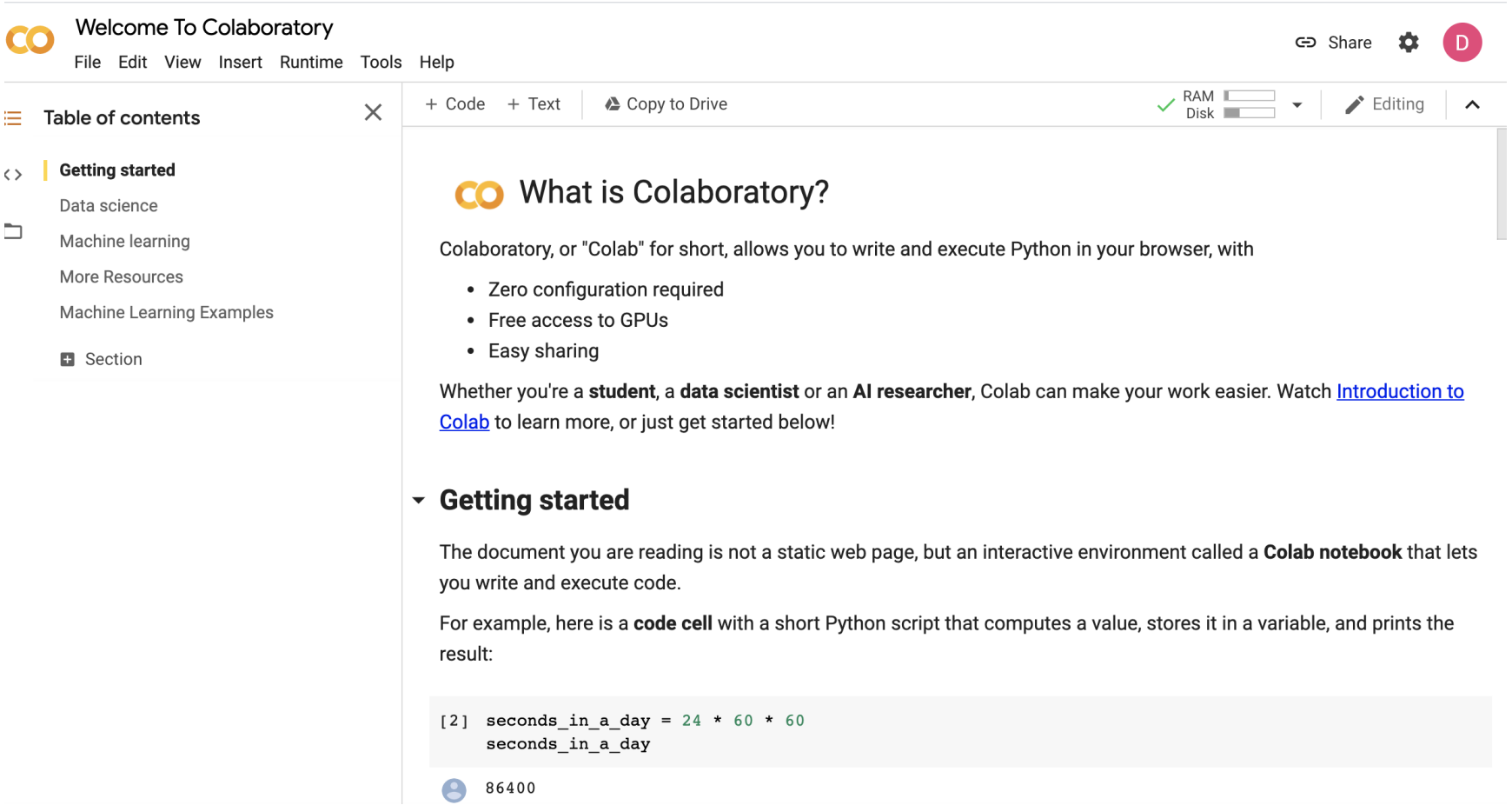
Freebuff privacy policy (effective November 12, 2026) (checked November 9, 2026)

Bring your own data (optional)

- Workshop example data will be available; bringing your own data is optional.
- Bring a small public or synthetic sample you are allowed to use, plus a question.
- Add a README / data dictionary: columns, units, provenance, and expected output.
- Do not upload patient-level, confidential, restricted, or unpublished data.
- Ask an instructor before using it; a private repository is not approval for AI use.

If your real data are sensitive, bring the schema or a synthetic example instead.

Inside a Colab notebook



Historical MLE 0426 Module 1 example. The following seven slides are original workshop slides; current Colab screens and features may differ.

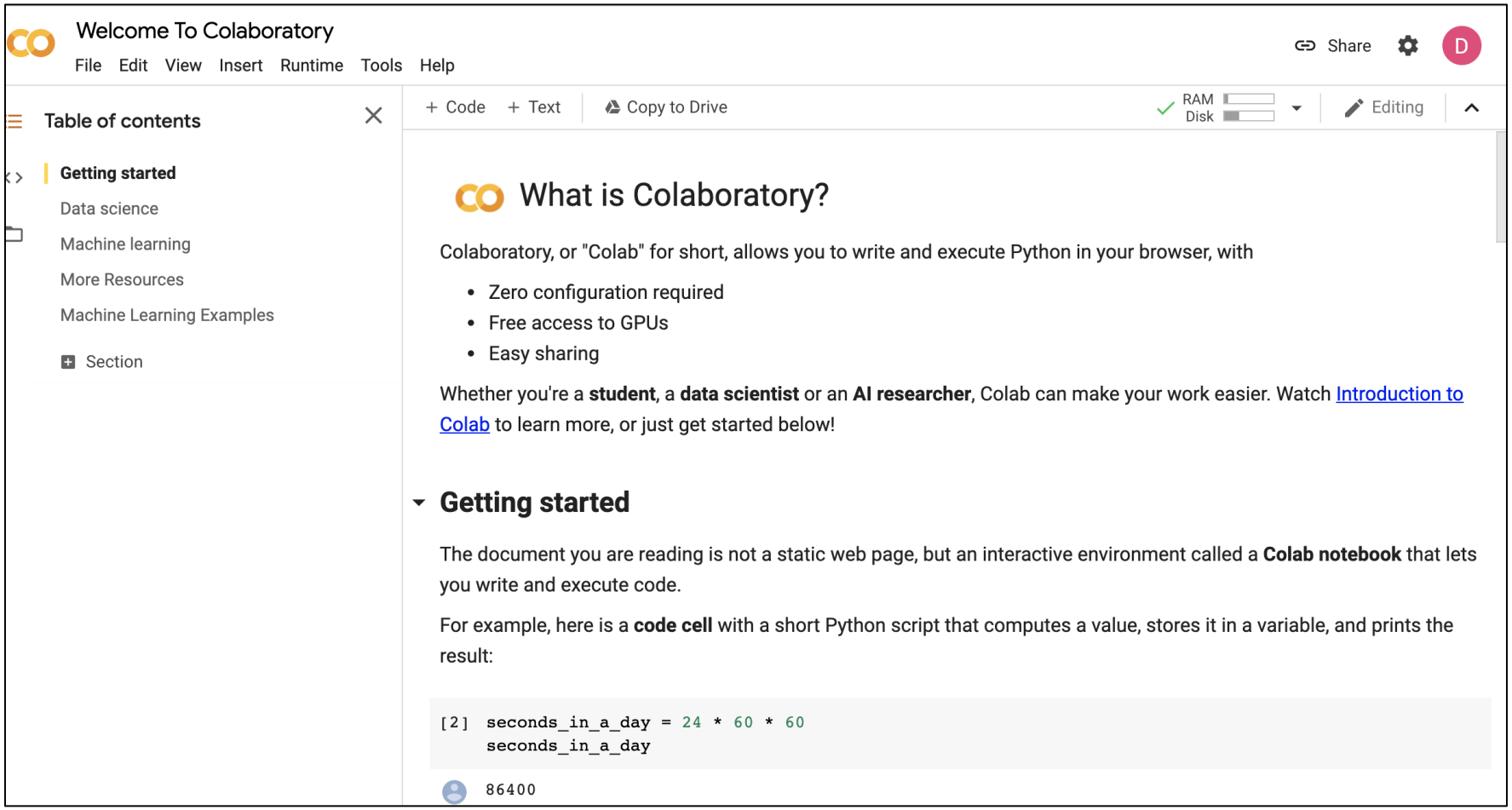
Google Colab

- Colab is a Python development environment that runs in the browser using Google Cloud
- Need to have a google account/google drive (see next slide)
- No need to install or run Python locally or deal with local OS issues
- Allows you to access, import and edit previously written Python code
- Allows you to share, save and download code
- Allows easy access to NumPy
- Allows easy access to GPUs, ML code and TensorFlow for ML applications

Google Colaboratory

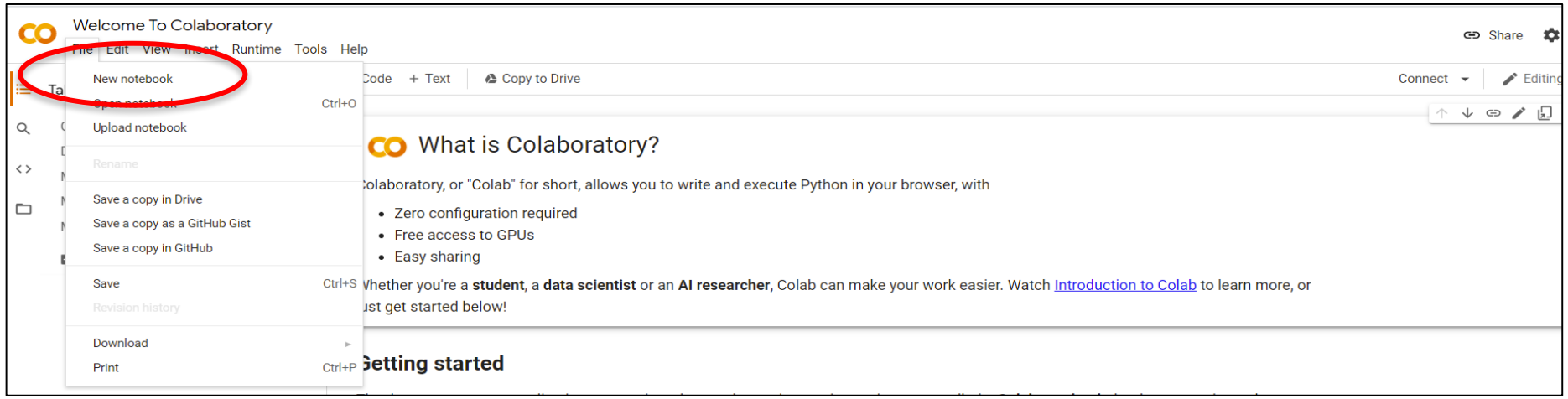
- Access by typing “Colab” in Google or go to https://colab.research.google.com/
- Click on “Welcome to Colaboratory” to launch
- Click on the “Introduction to Colab” video to learn more (3 minutes)
- Colab home page has File/Edit/View/Insert functions on upper left of page
- Runs like a Jupyter notebook
- Colab notebooks allow you to combine executable code, rich text, images, HTML, etc.

Colab Main Page



How to Open a New Notebook

- **Step 2: Under File, click New Notebook**
- **A new notebook should open up as shown below**





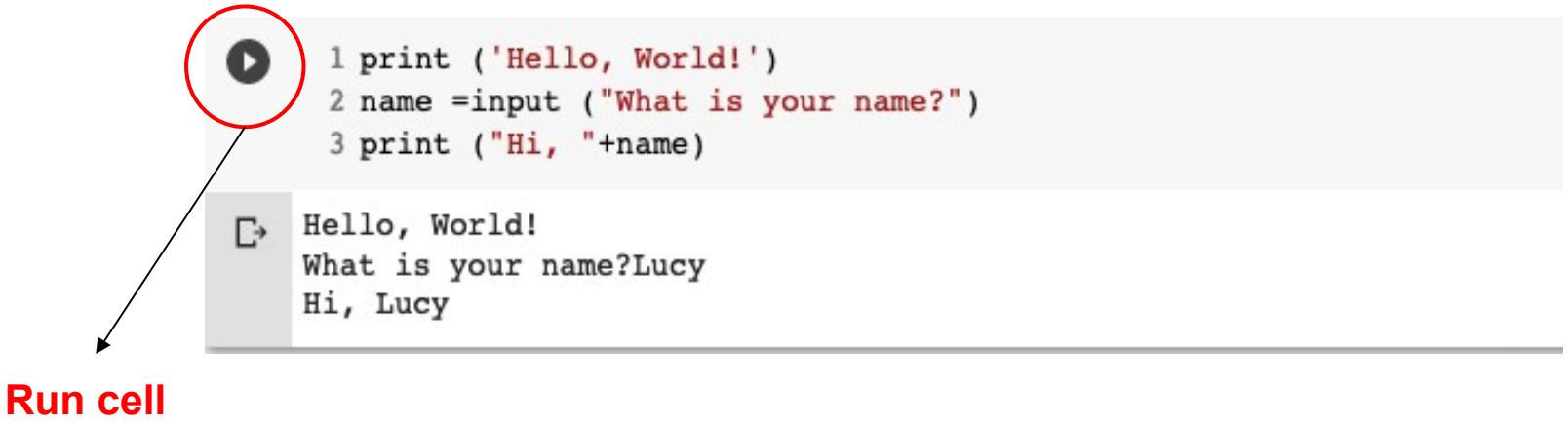
Hello World

- Let's write a program in Colab and call it "Hello World"
- Type that name in the circle and press Enter
- The notebook will acquire the name you have given now



Writing and Executing the Code

- Enter the trivial Python code and run it
- We will show how the simple code “Hello World” can be written
- Write this few lines of code in code cell
- Press run cell
- The output are the results



Runtime – Run All

- To save time, you can also click ‘Runtime’ then select ‘Run all’ to run all the cells

